

ELECTIVE COURSE-6.1

URA732: VEHICLE DYNAMICS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: The objective for this course is to introduce the concepts of vehicle dynamics where the students are exposed appropriate level of dynamic modeling that is required to analyze the performance of vehicle control systems. This understanding will include development of the analytical skills necessary to quantitatively predict the behavior of vehicle systems.				
Syllabus Introduction: Fundamental principles of vehicle dynamics, vehicle tires performance, cornering characteristics. Vehicle Kinematics: Fundamental principles of velocity, acceleration; two dimensional mechanisms. Multi-Body Systems Design: Three dimensional dynamics; dynamics of vehicle rollovers. Suspension Design: Passive, semi-active and active suspension. Vibrations Principles and Drive Dynamics: Drivetrain dynamics, vehicle performance. Steering Mechanisms: Two-dimensional analysis; mechanics of vehicle terrain interaction. Vehicle Collisions: Fundamental laws of motion; energy and momentum forces and moments.				
Laboratory Work (if applicable) 2D and 3D models of vehicles using MATLAB or Adams or bond graph for multibody dynamic analysis; Research assignments or projects.				
Course Learning Outcomes (CLOs) The students will be able to: 1. characterize the behaviour of vehicle systems and subsystems 2. develop models of vehicles for analysis of kinematics, (velocities and accelerations), kinetics (forces and moments). 3. perform simulations of rigid multi-body assemblies and calculation of loads, dynamic forces, energy and momentum in two and three dimensions.				
Text Books 1. Jazar, R., <i>Vehicle Dynamics</i>, Springer, (2008). 2. Thomas D., Gillespie., <i>Fundamentals of vehicle dynamics</i>, IEEE (1992). 3. Karnopp, Dean., <i>Vehicle Dynamics, Stability, and Control</i>, CRC Press, (2013).				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Reference Books

1. Wong, J.Y., *Theory of Ground Vehicles*, Third Edition, John Wiley (2001)
2. Matthew, Huang, *Vehicle Crash Mechanics*, CRC Press (2002)
3. Dixon John C., *Tyres, suspension and handling*, Cambridge University Press (1991)
4. Fenton, John, *Handbook of Vehicle Design Analysis*, SAE International (1996)

Evaluation Scheme:

Sl. No.	Evaluation Elements	Weightage (%)
1	MST	30
2	EST	45
3	Sessional (Assignments, Tutorials, Quizzes, Projects, Tests)	25

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